

A grayscale image of a 3D-printed titanium cage, showing its complex, lattice-like structure. The cage is oriented vertically, with its top and bottom flanges clearly visible. The background is a light gray with a subtle grid pattern.

Risk Factor for Cage Subsidence and Utility of Double 3D-printed Titanium Cage in Biportal Endoscopic Fusion

Comparison of 3-Dimensional-Printed Titanium and Polyetheretherketone Cages

Dong Hyun Lee

Department of Neurosurgery, Spine Center, Wiltse Memorial Hospital, Suwon

Choon Keun Park

Dong-Geun Lee

Jae Won Jang

Introduction

NEUROSURGICAL FOCUS

Neurosurg Focus 43 (2):E8, 2017

Fully endoscopic lumbar interbody fusion using a percutaneous unilateral biportal endoscopic technique: technical note and preliminary clinical results

*Dong Hwa Heo, MD, PhD,¹ Sang Kyu Son, MD,² Jin Hwa Eum, MD,³ and Choon Keun Park, MD, PhD,³ on behalf of the International Unilateral Biportal Endoscopic Spine Surgery Research Society

¹Department of Neurosurgery, Spine Center, The Leon Wiltse Memorial Hospital, Suwon; ²Department of Neurosurgery, Spine Center, Gangdong Hospital, Busan; and ³Department of Neurosurgery, Spine Center, Centum Hospital, Changwon, Korea

OBJECTIVE Minimally invasive spine surgery can minimize damage to normal anatomical structures. Recently, fully endoscopic spine surgeries have been attempted for lumbar fusion surgery. In this study, the authors performed a percutaneous unilateral biportal endoscopic (UBE) technique as a minimally invasive surgery for lumbar fusion. The purpose of this study is to present the UBE technique of fully endoscopic lumbar interbody fusion (LIF) and to analyze the clinical results.

METHODS Patients who were to undergo single-level fusion surgery from L3–4 to L5–S1 were enrolled. Two channels (endoscopic portal and working portal) were used for endoscopic lumbar fusion surgery. All patients underwent follow-up for more than 12 months. Demographic characteristics, diagnosis, operative time, and estimated blood loss were evaluated. MRI was performed on postoperative Day 2. Clinical evaluations (visual analog scale [VAS] for the leg and Oswestry Disability Index [ODI] scores) were performed preoperatively and during the follow-up period.

RESULTS A total of 69 patients (24 men and 45 women) were enrolled in this study. The mean follow-up period was 13.5 months. Postoperative MRI revealed optimal direct neural decompression after fully endoscopic fusion surgery. VAS and ODI scores significantly improved after the surgery. There was no postoperative neurological deterioration.

CONCLUSIONS Fully endoscopic LIF using the UBE technique may represent an alternative minimally invasive LIF surgery for the treatment of degenerative lumbar disease. Long-term follow-up and larger clinical studies are needed to validate the clinical and radiological results of this surgery.

<https://thejns.org/doi/abs/10.3171/2017.5.FOCUS17146>

KEY WORDS lumbar; fusion surgery; endoscopy; minimally invasive surgery

LUMBAR fusion surgery is a treatment option for lumbar degenerative disease, such as segmental instability.^{4,6,9,13,14} There are many types of lumbar fusion surgery, including anterior lumbar interbody fusion (ALIF), lateral lumbar interbody fusion (LLIF), posterior lumbar interbody fusion (PLIF), and transforaminal lumbar interbody fusion (TLIF).^{4,6,9,11} Among these treatment modalities, PLIF and TLIF can result in direct posterior neural decompression.¹¹ Although conventional posterior open fusion surgeries can achieve wide decompression of neural structures and provide stabilization for surgically

treated segments, they can also result in extensive destruction of posterior anatomical structures, thereby requiring long periods of recovery. Therefore, various types of minimally invasive spine surgeries have been attempted to treat lumbar degenerative disease.^{6,7,11,12,15,17} These surgeries can minimize injury to normal anatomical structures during the procedures.^{1,3,5,7,10} Recently, spinal endoscopic surgeries have been attempted in the field of lumbar decompression and fusion surgery.^{2,3,7,10,17}

We have been using the percutaneous unilateral biportal endoscopic (UBE) technique for lumbar discectomy,

Heo DH, Son SK, Eum JH, Park CK. Fully endoscopic lumbar interbody fusion using a percutaneous unilateral biportal endoscopic technique: technical note and preliminary clinical results. *Neurosurg Focus*. 2017;43(2): E8.

<https://doi.org/10.3171/2017.5.Focus17146>

ABBREVIATIONS ALIF = anterior lumbar interbody fusion; LLIF = lateral LIF; ODI = Oswestry Disability Index; TLIF = transforaminal LIF; UBE = unilateral biportal endoscopic; VAS = visual analog scale.

SUBMITTED March 4, 2017. **ACCEPTED** May 22, 2017.

INCLUDE WHEN CITING DOI: 10.3171/2017.5.FOCUS17146.

* Drs. Heo and Son contributed equally to this work.

©AANS, 2017

Neurosurg Focus Volume 43 • August 2017 1

Unauthenticated | Downloaded 05/08/24 11:25 AM UTC

Introduction

The unilateral biportal endoscopic TLIF was the major alteration to the endoscopic TLIF up to this point ¹⁷.

By only isolating the working portal from the endoscope, this method produces better ergonomics and is very comparable to the full-endoscopic uniportal approach. Compared to uniportal operations, special tools are not needed since the working and viewing portals are different and may be swapped as needed.

The biportal endoscopic technique is a revolutionary minimally invasive technique that abandons the tubular retractor.

The larger lens and the tool flexibility it provides are advantageous in complex spine surgeries that require detailed work on small structures. Surgeons can obtain different perspectives and magnifications without repositioning the instruments constantly.

It is performed through 2 independent portals with continuous irrigation of normal saline. The normal saline provides hydrostatic pressure to inhibit bleeding and carries away bone debris and oozing. The diameter of the endoscope is only 4 mm, thin enough to get access to the deep structures such as the contralateral lateral recess and the neural foramen. Combined with a high-definition endoscope, the biportal endoscopic technique provide a clear, bright, and magnified surgical field, enabling the surgeon to perform delicate surgical procedures with reduced risks of neural injuries.

Introduction

European Spine Journal (2023) 32:2853–2862
<https://doi.org/10.1007/s00586-023-07747-9>

ORIGINAL ARTICLE

Enhanced recovery after surgery pathway with modified biportal endoscopic transforaminal lumbar interbody fusion using a large cage. Comparative study with minimally invasive microscopic transforaminal lumbar interbody fusion

Dong Hwa Heo¹ · Jae Won Jang²  · Choon Keun Park²

Received: 21 December 2022 / Revised: 20 April 2023 / Accepted: 24 April 2023 / Published online: 21 May 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract
Purpose Studies about the clinical efficacy of endoscopic lumbar interbody fusion using an enhanced recovery after surgery (ERAS) pathway are insufficient. Thus, the purpose of this study was to investigate clinical usefulness of biportal endoscopic transforaminal lumbar interbody fusion (TLIF) using an ERAS compared with microscopic TLIF.
Methods Prospectively collected data were retrospectively analyzed. Patients who received modified biportal endoscopic TLIF with ERAS were grouped into an endoscopic TLIF group. Those who received microscopic TLIF without ERAS were grouped into a microscopic TLIF group. Clinical and radiologic parameters were compared between two groups. Fusion rate was evaluated using sagittal reconstruction images of postoperative computed tomographic (CT) scan.
Results There were 32 patients in the endoscopic TLIF group with ERAS and 41 patients in the microscopic TLIF group without ERAS. Visual analog scale (VAS) scores for back pain preoperatively at day one and day two were significantly ($p < 0.05$) higher in the non-ERAS microscopic TLIF group than in the ERAS endoscopic TLIF group. Preoperative Oswestry Disability Index were significantly improved at the last follow-up in both groups. The fusion rate at postoperative one year was 87.5% in the endoscopic TLIF group and 85.4% in the microscopic TLIF group.
Conclusion Biportal endoscopic TLIF with ERAS pathway may have good aspect to accelerate recovery after surgery. There was no inferiority of fusion rate of endoscopic TLIF comparing to microscopic TLIF. Biportal endoscopic TLIF using a large cage with ERAS pathway may be a good alternative treatment for lumbar degenerative disease.

Keywords Fusion · Lumbar · Endoscopy · Recovery · Biportal

Introduction

The proportion of the elderly population is increasing with the improvement of living standards and health care systems. Degenerative spinal diseases presenting with back pain and radicular pain are among the most frequently developed disease entities in the health care system. Recently, the number of patients with spinal diseases is rapidly increasing with

extended life expectancy, increasing the burden of an individual's life and human society [1]. Initial treatment option for a degenerative spinal disease is conservative treatment including bed rest, physiotherapy, pain killer medications, and pain procedures. Despite various conservative treatment options, some patients who have failed conservative management require surgical intervention to improve their quality of life.
Enhance recovery after surgery (ERAS) pathway is a widely accepted program in various surgical fields such as cancer surgery, orthopedic surgery, gynecologic surgery, abdominal surgery, and vascular surgery [2, 3]. Compared to other departments, the ERAS pathway might not be popular in the field of spinal surgery [3]. ERAS protocol usually consists of preoperative, intraoperative, and postoperative courses [2]. The most important part of ERAS may be the

 Springer

Heo DH, Park CK. Clinical results of percutaneous biportal endoscopic lumbar interbody fusion with application of enhanced recovery after surgery. *Neurosurg Focus*. 2019;46(4): E18.
<https://doi.org/10.3171/2019.1.Focus18695>

Neurospine 2023;20(1):80–91.
<https://doi.org/10.14245/ns.2346036.018>

Original Article

Biportal Endoscopic Transforaminal Lumbar Interbody Fusion Using Double Cages: Surgical Techniques and Treatment Outcomes

Jwo-Luen Pao^{1,2}

Received: January 14, 2023
Revised: February 28, 2023
Accepted: March 3, 2023


This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

Copyright © 2023 by the Korean Spinal Neurosurgery Society

INTRODUCTION

Lumbar interbody fusion (LIF) has been recognized as an effective surgical treatment for patients with refractory low back pain due to a variety of degenerative lumbar spinal disorders, including degenerative disc diseases and spondylolisthesis. LIF can be done via anterior or posterior approaches. The posterior approach is familiar to the spine surgeons and is capable of directly decompressing stenosis. Among the various LIF techniques via posterior approach, transforaminal lumbar interbody fusion

Objective: To describe the surgical techniques and the treatment outcomes of biportal endoscopic transforaminal lumbar interbody fusion (BETLIF) using double cages.
Methods: This study included 89 patients with 114 fusion segments between July 2019 and May 2021. One pure polyetheretherketone (PEEK) cage and 1 composite titanium-PEEK cage were used for interbody fusion. Clinical outcomes measures included visual analogue scale (VAS) scores for lower back pain and leg pain, Oswestry Disability Index (ODI), and Japanese Orthopedic Association (JOA) scores. Computed tomography (CT) of the lumbar spine 1 year postoperatively was used to evaluate the Bridwell interbody fusion grades.
Results: There were significant improvement in VAS for lower back pain from 5.2 ± 3.1 to 1.7 ± 2.1 , VAS for leg pain from 6.3 ± 2.5 to 1.7 ± 2.0 , ODI from 46.7 ± 17.0 to 12.7 ± 16.1 , and JOA score from 15.6 ± 6.3 to 26.4 ± 3.2 . The p-values were all < 0.001 . The average hospital stay was 5.7 ± 1.1 days. The CT studies available for 60 fusion segments showed successful fusion (Bridwell grade I or grade II) in 56 segments (93.3%). Significant cage subsidence of more than 2 mm was only noted in 3 segments (5.0%). Complications included 1 dural tear, 2 pedicle screws malposition, and 2 epidural hematomas, in which 2 patients required reoperations.
Conclusion: BETLIF with double cages provided good neural decompression and a sound environment for interbody fusion with a big cage footprint, a large amount of bone graft, endplate preservation, and segmental stability.

Keywords: Minimally invasive surgery, Biportal endoscopy, Interbody fusion, Fusion cage, Computed tomography, Treatment outcomes

 www.e-neurospine.org

Pao JL. Biportal Endoscopic Transforaminal Lumbar Interbody Fusion Using Double Cages: Surgical Techniques and Treatment Outcomes. *Neurospine*. 2023;20(1): 80–91.
<https://doi.org/10.14245/ns.2346036.018>

Neurospine 2023;20(1):110–118.
<https://doi.org/10.14245/ns.2346116.058>

Original Article

The Use of Dual Direction Expandable Titanium Cage With Biportal Endoscopic Transforaminal Lumbar Interbody Fusion: A Technical Consideration With Preliminary Results

Don Young Park¹, Dong Hwa Heo²

Received: January 25, 2023
Revised: February 27, 2023
Accepted: February 27, 2023


This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

Copyright © 2023 by the Korean Spinal Neurosurgery Society

INTRODUCTION

Minimally invasive transforaminal lumbar interbody fusion (MIS-TLIF) has demonstrated comparable clinical outcomes and safety profile as compared to open conventional TLIF with significant improvement of pain and disability.^{1,2} More recently, endoscopic techniques to perform TLIF surgery have been introduced with similar success as MIS-TLIF, especially with biportal endoscopic techniques.^{3,4} The biportal endoscopic TLIF

Objective: Expandable cage technology has emerged for lumbar interbody fusion to restore intervertebral disc space height and alignment through a narrow surgical corridor. The purpose of this study is to present the technique of biportal endoscopic transforaminal lumbar interbody fusion (TLIF) using dual direction expandable cage and provide early clinical results.
Methods: We performed the biportal endoscopic TLIF using a dual direction expandable titanium cage for height restoration and a larger footprint in 10 patients. Clinical parameters including Oswestry Disability Index (ODI), visual analogue scale (VAS), and complications were retrospectively analyzed. Also, we investigated radiologic parameters using preoperative and postoperative x-ray images.
Results: We successfully inserted dual direction expandable cages during biportal endoscopic TLIF. There was no significant subsidence or collapse of the expandable cages during the 6-month follow-up period. Lumbar lordosis and disc height were significantly increased after surgery. ODI and VAS scores were significantly improved at 6 months after surgery.
Conclusion: In this report, we describe the first use of a dual direction expandable interbody TLIF cage that expands in both width and height in biportal endoscopic TLIF surgery. Early clinical and radiographic outcomes of this TLIF technique may be favorable in early 6-month follow-up.

Keywords: Endoscopy, Lumbar vertebrae, Surgery, Biportal

 www.e-neurospine.org

Park DY, Heo DH. The Use of Dual Direction Expandable Titanium Cage With Biportal Endoscopic Transforaminal Lumbar Interbody Fusion: A Technical Consideration With Preliminary Results. *Neurospine*. 2023;20(1): 110–118.
<https://doi.org/10.14245/ns.2346116.058>

Object

The biportal endoscopic technique is a revolutionary minimally invasive technique that abandons the tubular retractor.

Recently, the biportal endoscopy technique were applied to lumbar interbody fusion surgery in several pioneer studies.

Currently, various cages are used in endoscopic fusion.

Efforts are being made to increase the fusion rate and prevent subsidence, but research using **double 3-Dimensional-printed titanium cage has not been conducted**, and we would like to report an optimized cage of endoscopic fusion by comparing the double cage made of PEEK.

Methods

Between January 2020 and April 2023, we retrospectively analyzed 89 patients who underwent biportal endoscopic transforminal lumbar interbody fusion (BETLIF) (*3D-printed titanium, 48 levels; PEEK, 47 levels*) in two spine facility.

Radiologic parameters were assessed with x-ray and computed tomography (CT) at 6 months, and 1 year.

Clinical status was evaluated using Oswestry Disability Index, visual analogue scale score, and Bridwell fusion grading was assessed on 6-month and 1-year postoperative CT.

Cage subsidence was considered clinically relevant as a significant subsidence of 2 mm or more in the depth at which the cage migrated into the endplate on sagittal or coronal CT images.

Materials and methods

1 *Subjects*

1. Spine Surgery Research Data

- Patients with spine conditions underwent BETLIF surgeries at two facilities.
- Forty-six patients had a 3D-printed titanium cage while forty-three had a PEEK cage.
- Patients continued to experience pain despite initial conservative therapy.
- **No anabolic medications were administered to the patients.**
- The study focused on perioperative, clinical, and radiologic data of patients undergoing BETLIF.

Materials and methods

2 Preparation of 3D-Printed Titanium Cage

Medical Device Implementation

- The patient's disc size was accommodated with a Lumbar 3D PLIF titanium cage from Genoss in Korea. (Fig. 1)
- The 3D model was created using Selective Laser Melting technology on an EOSINT-M280 3D printer. (EOSINT-M280, EOS GmbH, Munich, Germany).²²
- The 3D-printed titanium cage features a mean pore size range of 1100 μm and mean porosity range of 70%. Both PEEK and the titanium cage utilized a straight bullet type, considering the patient's anatomical features for measurements. (Request confirmation from genoss)

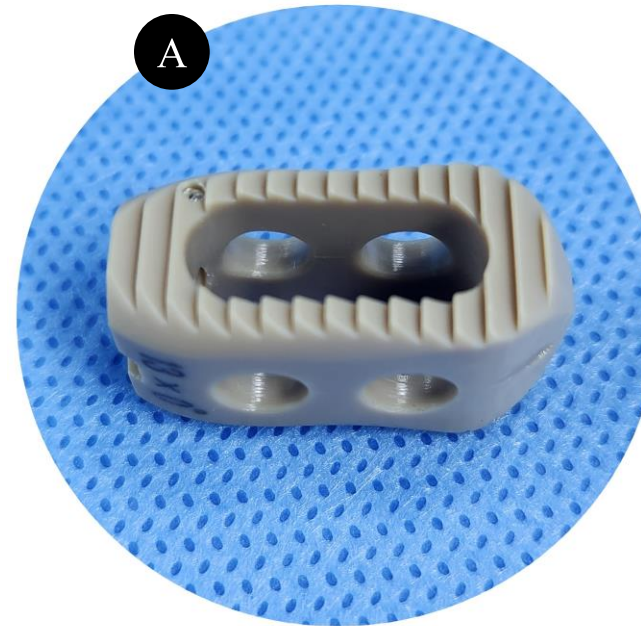
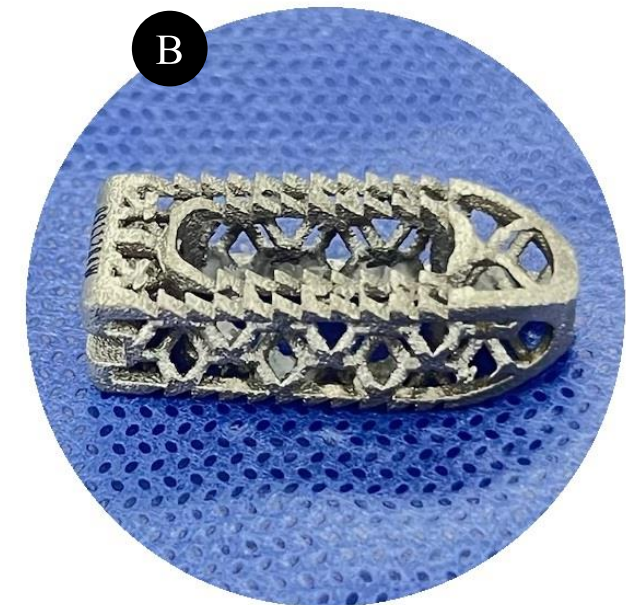


Figure 1. Interbody cages.

A polyetheretherketone cage (IVA PLIF cage, Huvexel, Seongnam-si, Kyunggi-do, Korea) (A) and a 3-dimensional (3D)-printed titanium cage (Lumbar 3D PLIF, Genoss, Suwon, Kyunggi-do, Korea) (B). The 3D-printed cage had an internal lattice-like structure.



Evaluation of the radiological and clinical results

The patients' records that were included were assessed.

Six months and a year following surgery, respectively, the CT and x-ray scans were carried out. Based on CT scans, the Bridwell fusion grading system was used to identify bone fusion ix months and a year after surgery.²³

The Bridwell fusion grades were as follows : grade I, which was fused with the formation of trabecular bone; grade II, which was intact but not entirely remodelled; grade III, which was intact but had lucency above and below the graft; and grade IV, which was devoid of fusion with graft collapse or resorption (Fig. 2).

The total of grades I and II was the computed fusion rate.

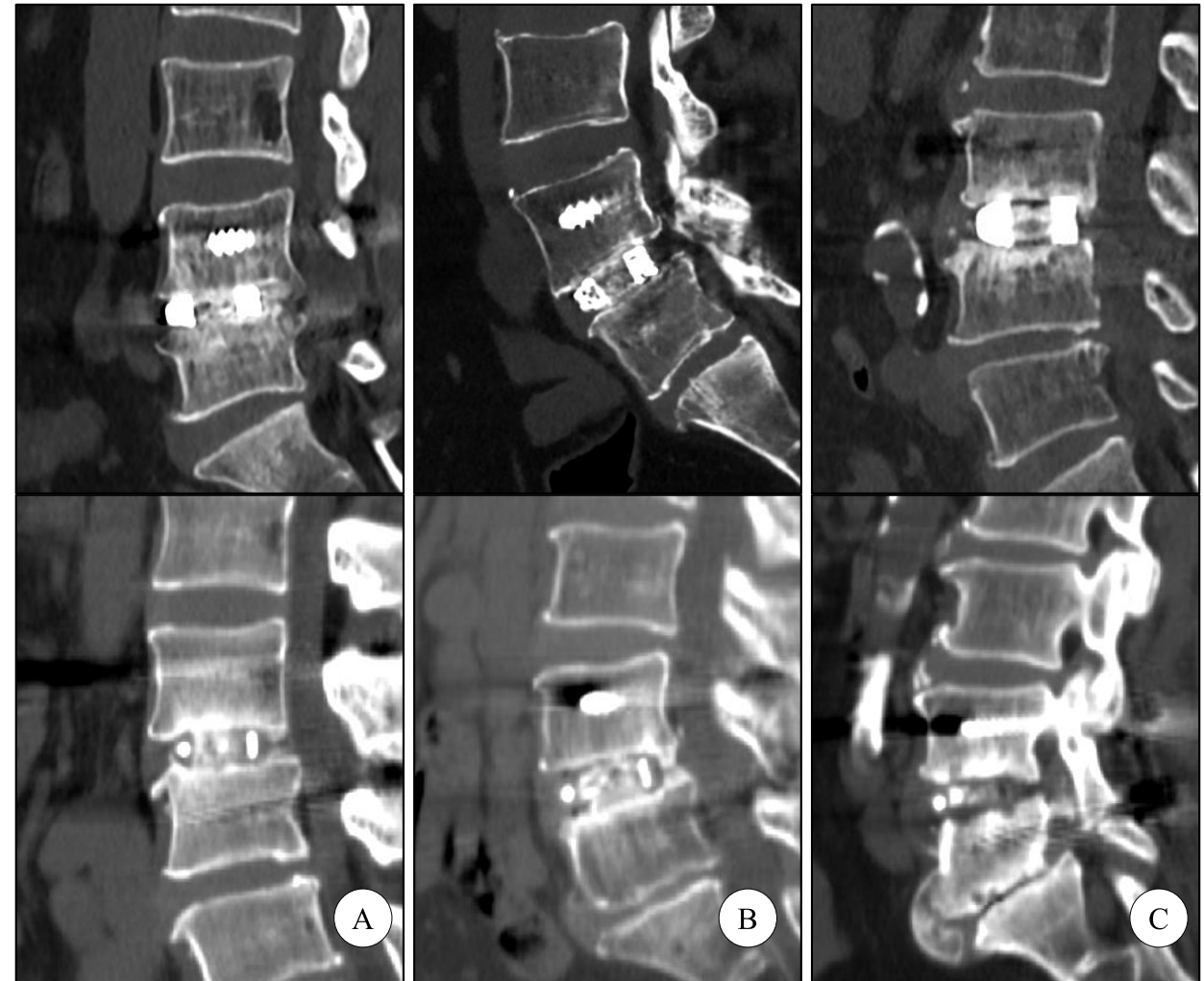


Figure 2. Interbody cages. A polyetheretherketone cage (CAPSTONE, Medtronic, Memphis, TN, USA) (A) and a 3-dimensional (3D)-printed titanium cage (PANTHER, Mantiz, Daegu, Korea) (B) The 3D-printed cage had an internal lattice-like structure.

Evaluation of the radiological and clinical results

Medical Data Analysis

- Confirm subsidence grade using CT data at 1-year point.
- Analyze dual-energy x-ray absorptiometry results(DEXA).
- Measure cage dimensions, history of tobacco usage and BMI.
- Utilize demineralized bone matrix putty for graft material inside and outside the cages.

Cage Subsidence Classification

- Cage subsidence classified by cage migration depth into endplate on CT images.
- Classified as 'no more than 2 mm,' or '2 mm or more.'
- 2mm, more than 2 mm subsidence deemed clinically significant. ^{24,25}

Surgical technique of biportal endoscopic TLIF using a Double cage

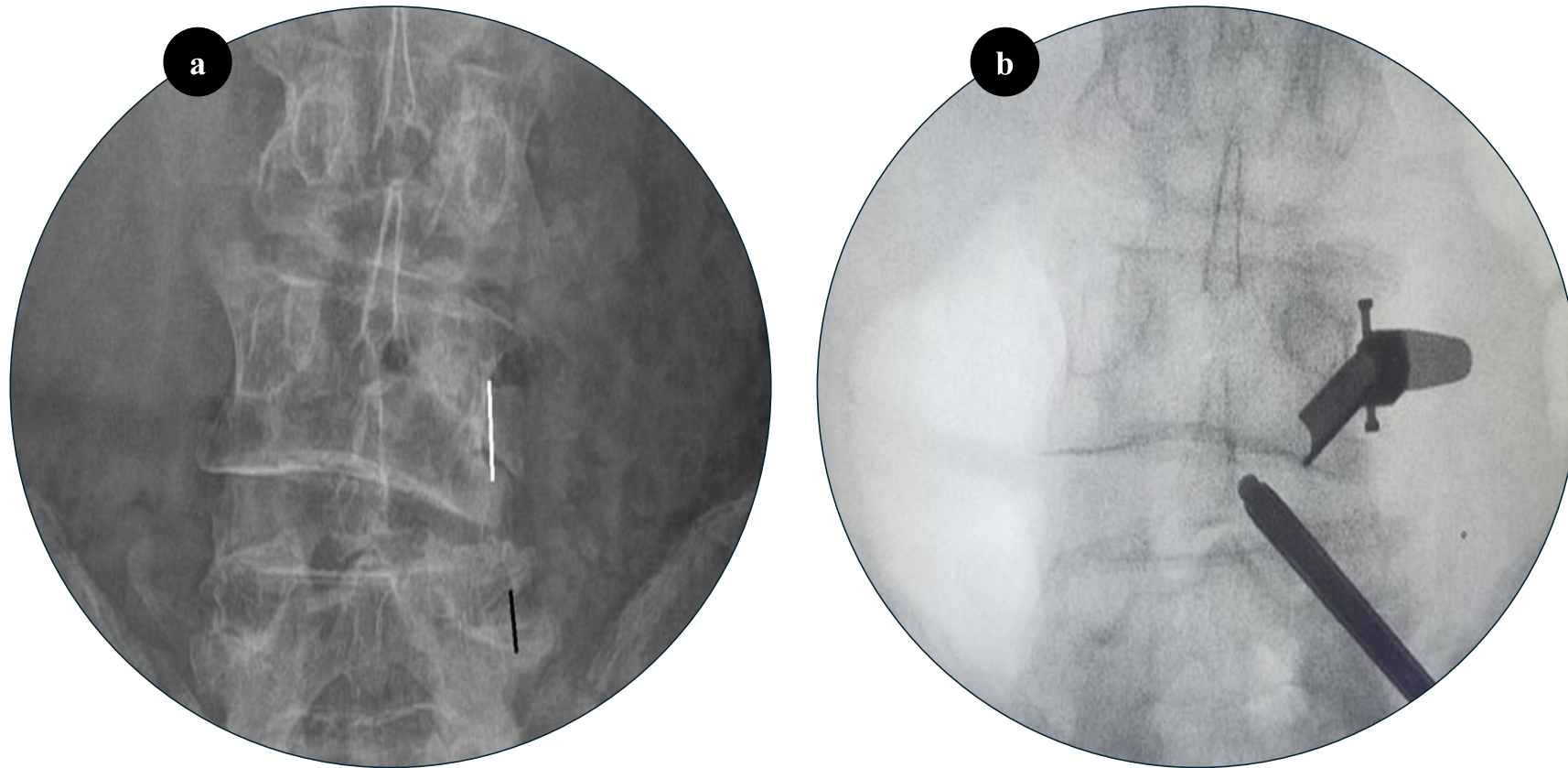


Figure 3. Skin incision points for biportal endoscopic transforaminal lumbar interbody fusion

a Black line is skin incision point for an endoscopic viewing portal, and white line is skin incision point for a working portal. C-arm fluoroscopic X-ray image after making two portals **b** A working sheath was inserted into working portal

Surgical technique of biportal endoscopic TLIF using a Double cage

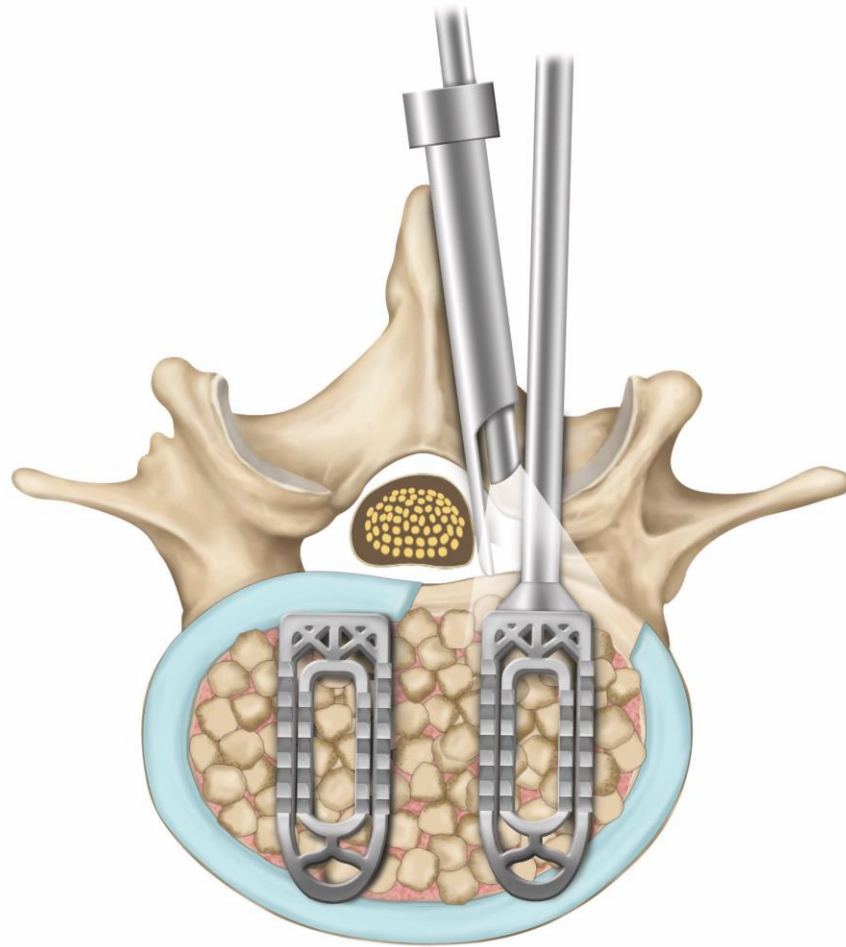
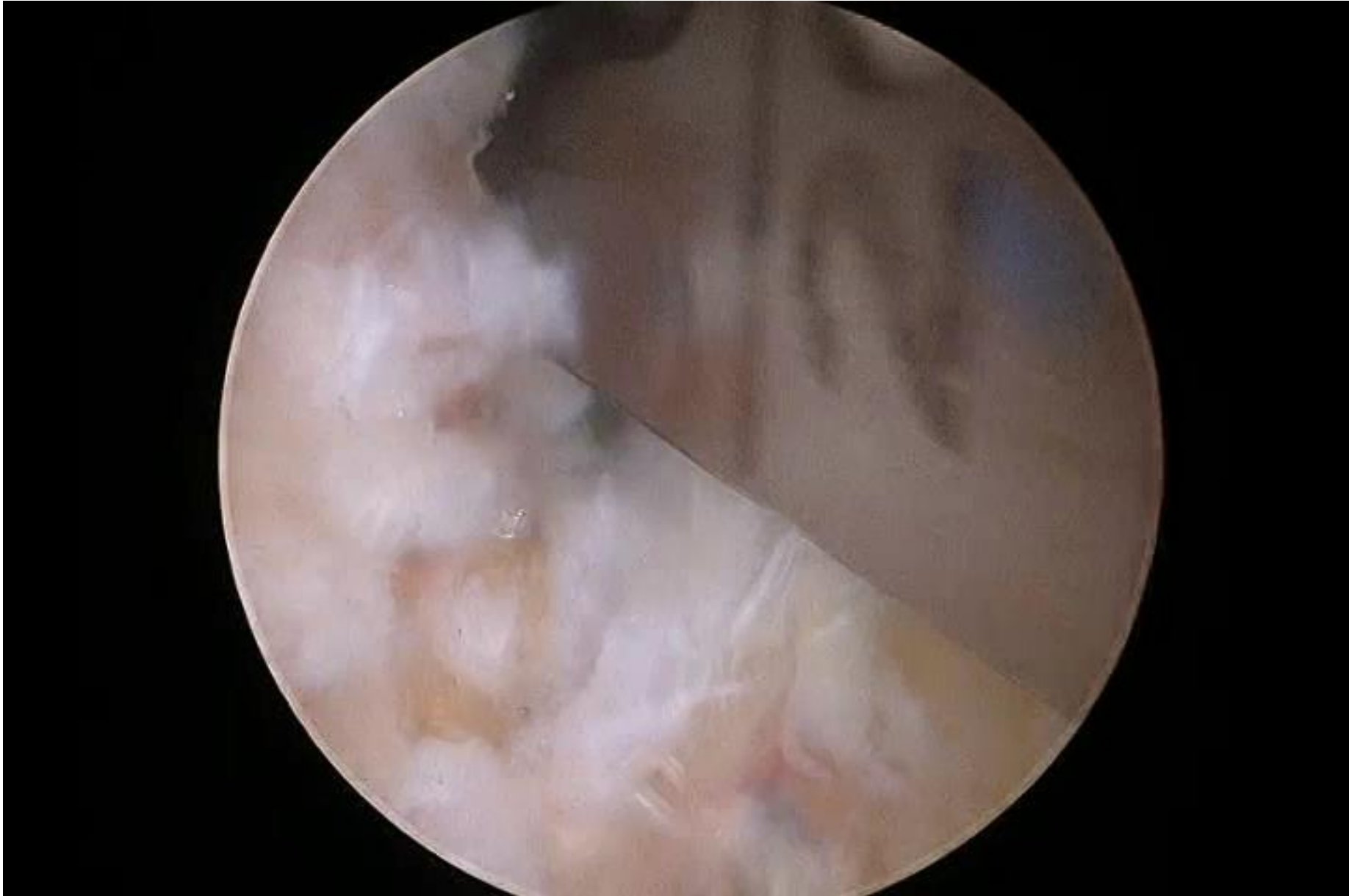


Figure 6B. (A) The cages and their dimensions. (B–D) Illustrations and endoscopic photos demonstrate vertical insertion of the first cage, impaction of bone graft into the disc space maintained by the first cage, and 45° oblique insertion of the second cage.

Surgical technique of biportal endoscopic TLIF using a Double cage



Surgical technique of biportal endoscopic TLIF using a Double cage

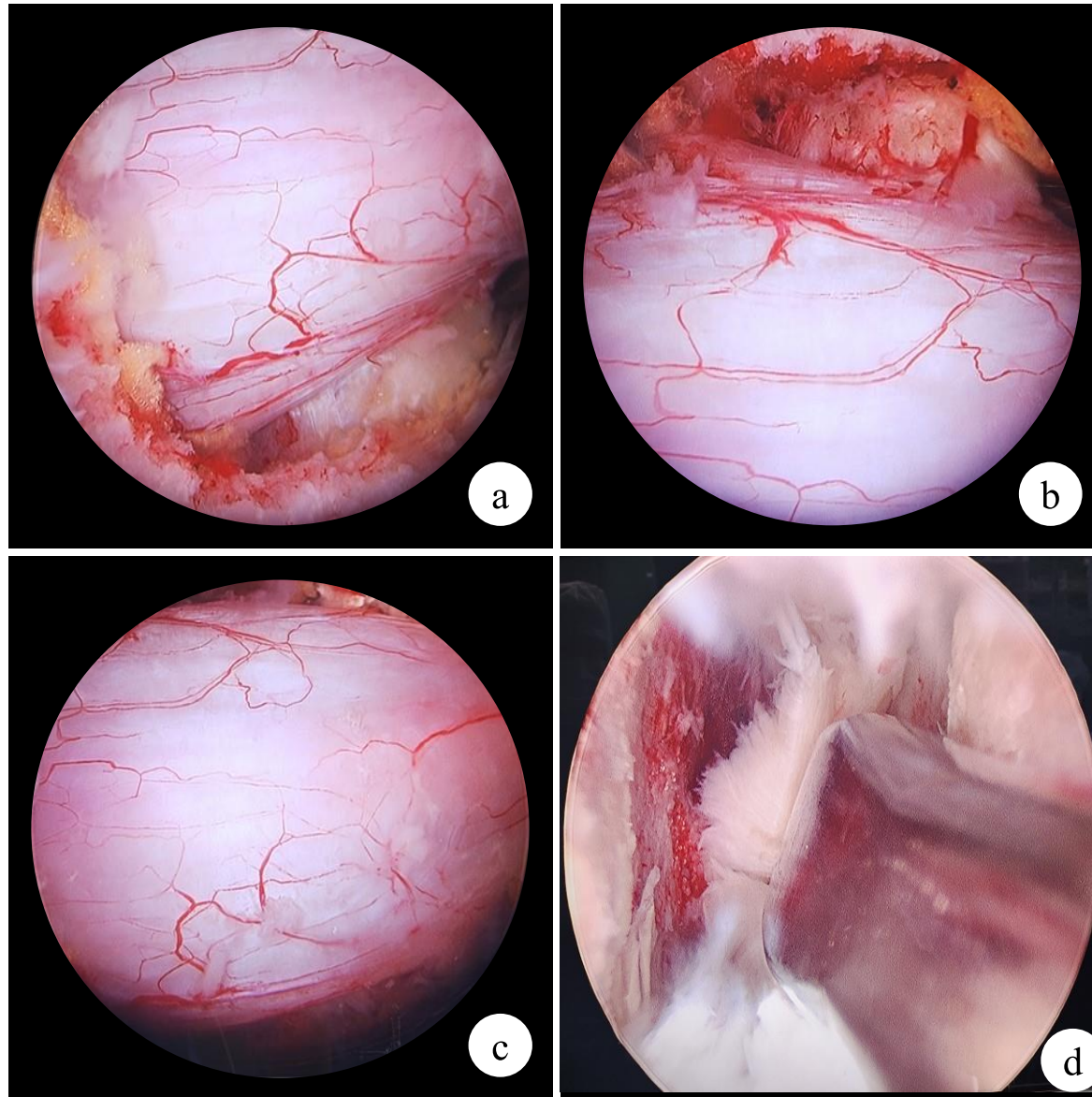


Figure 4. Intraoperative endoscopic images of modified biportal endoscopic transforaminal lumbar interbody fusion. Fully decompressive status of left ipsilateral traversing nerve root **a** Contralateral traversing nerve root **b** Central canal **c** Endoscopic endplate preparation image **d** cartilaginous endplate is separated from osseous endplate using a dissector

Surgical technique of biportal endoscopic TLIF using a Double cage

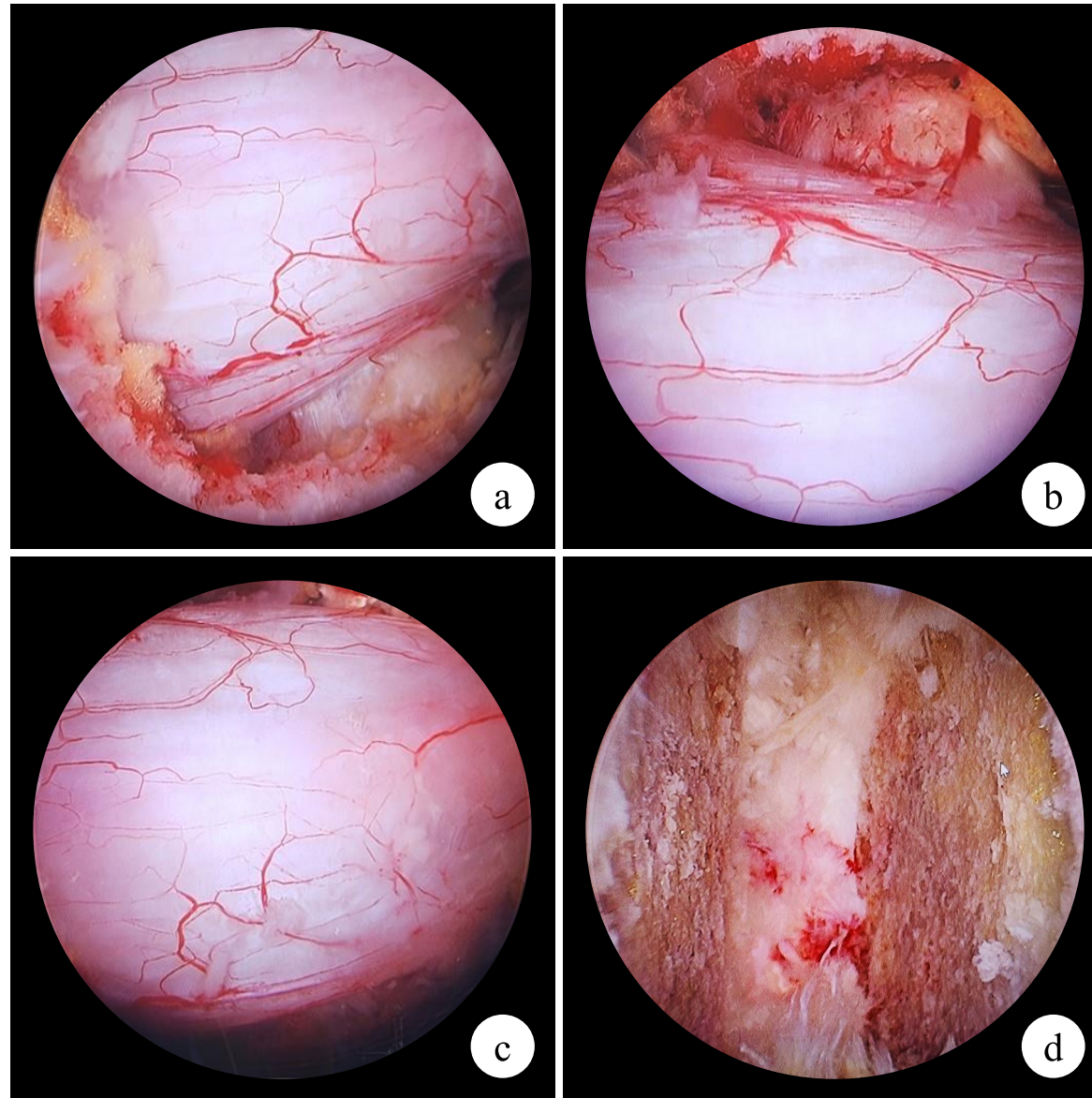


Figure 4. Intraoperative endoscopic images of modified biportal endoscopic transforaminal lumbar interbody fusion. Fully decompressive status of left ipsilateral traversing nerve root **a** Contralateral traversing nerve root **b** Central canal **c** Endoscopic endplate preparation image **d** cartilaginous endplate is separated from osseous endplate using a dissector

Results

Table 2. Group comparisons of demographic and clinical data

Variable	3D-Titanium	PEEK	P-Value
Number of patients	46	43	
Number of spinal fusion levels	48	47	
Sex ratio (females:males)	36:10	27:16	0.109
Age (years, mean±SD)	65.6	66.4	0.576
Smokers, n	4	3	0.763
BMI(kg/m ²)	25.57±3.21	24.91±3.01	0.354
BMD (T-score)	-1.48 ± 0.94	-1.44 ± 0.94	0.843
Osteopenia/osteoporosis, n (%)	34 (70.8)	32 (68.1)	0.771
Cage length in mm, % implanted			
22	10		
24	20		
26	14	47	0.001
28	1		
30	3		
Follow-up (weeks)	71.1	67.5	
Clinical results			
Preop ODI (%)	50.53±23.64	48.58±14.21	0.544
Postop ODI (%)	37.43±13.37	36.86±8.77	0.782
Preop VAS (back pain)	4.62±1.46	4.82±0.91	0.576
Postop VAS (back pain)	2.22±0.91	2.40±1.26	0.245
Preop VAS (leg pain)	7.67±1.86	7.75±2.83	0.842
Postop VAS (leg pain)	2.42±2.22	2.82±1.88	0.834

Values are presented as number (%) or mean±standard deviation. 3D, 3-dimensional ; PEEK, polyetheretherketone ; BMD, bone mineral density ; BMI, body mass index; ODI, Oswestry Disability Index ; VAS, visual analogue scale. Chi-square test, Student t-test, and Mann-Whitney U-test were used for statistical analysis.

Results

Table 3. Radiologic parameter.

Variable	Titanium	PEEK	p Value
Fusion Grade (Bridwell fusion grade) at 6-month follow-up			
Grade I	12 (25.0)	8 (17.0)	0.34
Grade II	30 (62.5)	31 (66.0)	0.725
Grade III	6 (12.5)	8 (17.0)	0.534
Grade IV	0	0	
Fusion rate (grade I + grade II)	42 (87.5)	39 (83.0)	0.534
Fusion Grade (Bridwell fusion grade) at 12-month follow-up			
Grade I	36 (75.0)	25 (53.2)	0.027*
Grade II	10 (20.8)	19 (40.4)	0.038*
Grade III	2 (4.2)	3 (6.4)	0.629
Grade IV	0	0	
Fusion rate (grade I + grade II)	46 (95.8)	44 (93.6)	0.629
Subsidence†	12 (25.0)	10 (21.3)	0.667

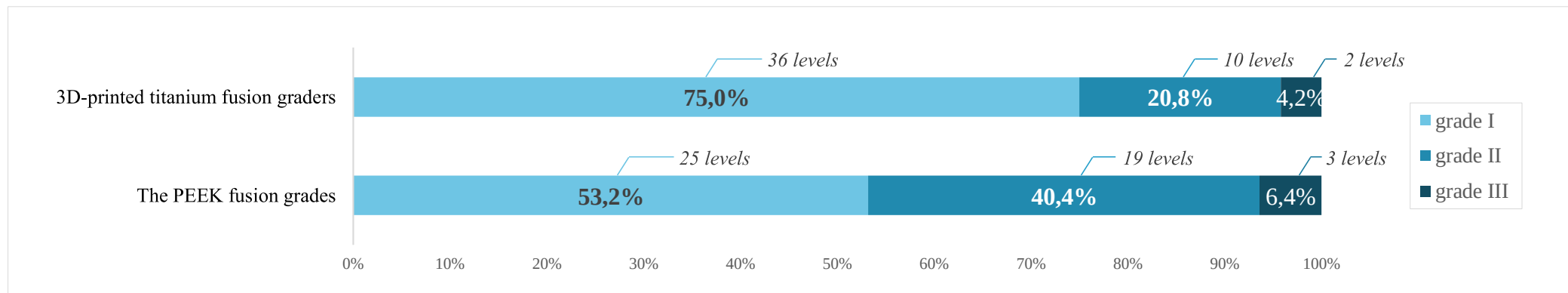
Values are presented as number (%) or mean±standard deviation. * $p < 0.05$, statistically significant difference. †Subsidence is defined segmental vertebral body height decreased by more than 2 mm at last follow-up after surgery than at immediate postoperative x-ray. Chi-square test, Student's t-test, and Mann-Whitney U-test were used for statistical analysis.

Results

There were no differences between the 2 groups in demographics and clinical outcomes.

At 1-year of follow-up, the reported **3D-printed titanium fusion grades** were grade I : 75.0% (36 levels), grade II : 20.8% (10 levels), and grade III : 4.2% (2 level).

The PEEK fusion grades were grade I : 53.2% (25 levels), grade II : 40.4% (19 levels), and grade III : 6.4% (3 levels).



For overall fusion rate (grade I, II), there was no difference between the 2 cages (95.8% vs. 93.6%, $p=0.629$), but grade I was reported at a higher incidence in 3D-printed titanium than PEEK (75.0% vs. 53.2%, $p=0.027$). There was no difference between cages based on subsidence and complications. Multivariate analysis with age, peek cage use, and bone mineral density (BMD T score ≤ -2.5) as variables showed that age is the only variable that significantly corresponds with a binarized subsidence in BETLIF.



Results

1. Comparison of Subsidence Rates

- **12 levels (25.0%)** subsidence in 3D-printed titanium cage vs. **10 levels (21.3%)** in PEEK cage.
- 3D-Titanium exhibited significantly higher subsidence rate at 12-month follow-up.

2. Factors associated with subsidence in BETLIF

- **Poor bone mineral density (BMD T score ≤ -1.0), (BMD T score ≤ -2.5)** and **older** patient age correlate with higher subsidence at 1-year follow-up.
- Osteopenia or/and osteoporosis usage significantly linked to subsidence at 12-month follow-up.
- P values of 0.050, 0.017 and 0.010 for BMD T score (BMD T score ≤ -1.0), (BMD T score ≤ -2.5) and age, respectively.

Age is the only variable that significantly corresponds with a binarized subsidence, according to multivariate analysis (Table 4).

This finding showed a linked increase in the probability of subsidence by factors of 1.08 for 12-month periods, respectively, with every extra year of age.

Furthermore, no greater sinking in the sample was noted in comparison to nonsmokers' subsidence.

Results

Table 4. Univariate analysis of subsidence at 12 months.

characteristic	group	Subsidence		p
		no	yes	
		N(%) M±SD	N(%) M±SD	
sex	male	20 (76.9)	6 (23.1)	.941
	female	48 (76.2)	15 (23.8)	
materials	PEEK	37 (78.7)	10 (21.3)	.667
	3D-titanium	36 (75.0)	12 (25.0)	
Smoking	smoking	5 (71.4)	2 (28.6)	.747
	non smoking	63 (76.8)	19 (23.2)	
BMD (T score≤-1)	-1 or less	47 (71.2)	19 (28.8)	.050
	-1 exceeds	26 (89.7)	3 (10.3)	
Osteoporosis (BMD T score≤-2.5)	-2.5 or less	10 (55.6)	8 (44.4)	.017
	-2.5 exceeds	63 (81.8)	14 (18.2)	
age		64.71 ± 8.01	70.10 ± 6.97	.010
Cage depth		25.45 ± 1.79	25.36 ± 1.05	.483
BMI(kg/m2)		25.22 ± 3.07	25.33 ± 3.32	.602



Results

Table 5. Multivariate analysis of subsidence at 12 months.

Follow-up interval	OR	Upper CI	Lower CI	p Value
12 months				
Age	1.077	1.158	1.002	0.044*
PEEK usage	1.003	3.040	0.331	0.995
Osteoporosis (BMD T score ≤ -2.5)	2.684	8.932	0.806	0.108
Cage depth	0.944	1.312	0.679	0.733

B : Unstandardised Beta, SE : Standard Error, OR : Odds ratio

Discussion

Advantages of Endo-TLIF

- Endo-TLIF minimizes postoperative morbidities like soft tissue trauma and blood loss.
- It also speeds up recovery, increases safety, improves efficacy, and delivers good outcomes.
- Uniportal endoscopic procedures are technically complex compared to biportal methods.
- The method's recognition has grown due to these benefits

Endplate preparation in BETLIF

- Endplate damage can be higher in osteoporosis patients, BETLIF has the potential to subvert this risk.
- The cartilaginous endplate is delicately peeled from the bony endplate in BETLIF using a freer elevator.
- Utilizing traditional tools like disc shavers and curettes could lead to endplate damage.
- Endoscope observation prevents endplate damage in UBE endoscopic fusion.
- Regular endoscope checks ensure complete disc removal with well-preserved bony endplate.
- The effectiveness of sophisticated UBE fusion in reducing endplate tasks is debatable.

Discussion

Spinal Fusion Two cage Techniques and Results

- Enhanced segmental stability with larger cage footprint reduces the risk of sinking.
- Inserting a large-sized cage was limited at the upper lumbar spine , more caution is required due to the risk of nerve root damage.
- The double cage insertion method lowers the risk of neurologic damage or postoperative neurological problems that may occur due to over retraction caused by insertion of a large cage.
- In our investigation, more bone graft may be affected in the disc space thanks to the sequential insertion of the double cages. two cage technology provides a large cage floor area.
- By increasing the contact between the exposed bony endplates and the bone graft, a significant volume of bone graft facilitates bone fusion. Additionally, after a successful fusion, it offers strong anterior support.
- The UBE double 3D-titanium cage can provide a high fusion rate of 95.8% in BETLIF.

Limitations

1. Study Limitations and Results

- The study faced limitations due to surgeon-specific biases and small sample size.
- Age was the only factor yielding a statistically significant outcome in multivariate analysis.
- Insufficient sample size may hamper the transfer of findings to the multivariate analysis.

2. Factors affecting surgical procedures.

- The BETLIF requires a complicated set of duties throughout the operational phase, and there are no clinical trials that provide a consistent process mapping for this treatment.
- Age, patient comorbidities, surgeon expertise, equipment, and implant materials, and case complexity affect the intraoperative surgical procedure.
- Incorrect endplate preparation for BETLIF can lead to graft subsidence.
- Sinking rates and subsidence are influenced by factors such as distraction, age, bone density, cage design, and cage width.

Conclusions

Double 3D-printed titanium cages showed better early fusion grades and comparable subsidence to that of using PEEK cages. The strongest indicator of subsidence was age in BETLIF.

Subsidence in BETLIF is probably not a binary outcome of cage selection, but rather the product of a multivariate process. These findings suggest that double 3D-printed titanium cages offer advantages in fusion quality, while further research is needed to optimize cage design, cage arrangement and surgical techniques to improve outcomes in biportal endoscopic fusion.

Thank you for your attention

Department of Neurosurgery, Spine Center, The Leon Wiltse Memorial Hospital

Dong Hyun Lee

